IRoC-U 2026

ISRO Robotics Challenge - URSC

Let's build a space robot

Proposal Report

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1. Description of ASCEND

1.1 Concept Overview

The proposed ASCEND (Autonomous Surveyor for Exploration, Navigation and Dynamics) is a battery-powered micro quadrotor designed for fully autonomous survey operations in a GPS-denied planetary-analog arena. The system integrates a high-agility rotorcraft airframe, a dedicated flight controller for low-level attitude and thrust control, and an onboard companion computer for perception, SLAM, feature detection, and mission planning. The ASCEND operates from a fixed base station that provides docking, battery charging, and high-bandwidth data offload via a wired and wireless interface, enabling multiple sorties within the stipulated mission time.

The overall objective is to emulate a planetary aerial scout that can:

- Acquire seed images of scientifically relevant surface features at the base.
- Perform systematic autonomous search over the arena.
- Detect and localise instances of those features.
- Return, dock, and transfer data without human intervention.

1.1 Why we choose Quadcopter

- The ASCEND platform is a micro quadcopter: a four-rotor UAV in X-configuration with fixed-pitch props. Opposite motors spin CW and CCW to balance torque; we tweak individual RPMs for net thrust and roll/pitch/yaw control. This setup gives direct authority over three rotations (roll, pitch, yaw) and vertical translation via differential thrust.
- The quadcopter is **under-actuated** in the sense that it has four actuators (rotors) but six degrees of freedom (3D position + 3D attitude). Position control in the horizontal plane is achieved indirectly: the flight controller tilts the vehicle to generate a horizontal component of thrust, and then regulates altitude and attitude using feedback from its inertial sensors (IMU) and barometer. Due to its symmetric structure and low rotational inertia, the quadcopter exhibits high agility and fast response, which is advantageous for precise low-speed manoeuvres, waypoint tracking, and hover in constrained environments.
- Control stack is cascaded: fast inner attitude loops (PID or model-based) stabilize angles, outer loops handle position/velocity. PX4 off-the-shelf stacks nail this on COTS flight controllers, shrugging off wind gusts and param tweaks. Leaves us free to bolt on SLAM and perception without reinventing hover.

1.2 Explanation for selecting the quadcopter

- A quadcopter configuration was selected due to its optimal balance between structural simplicity, mass efficiency, stability, and control authority for micro-UAV operations relevant to the ASCEND challenge.
- Compared to hexarotor systems, quadcopters require fewer motors and ESCs, resulting in lower all-up weight, reduced power consumption, and simpler wiring, which is critical for accommodating onboard perception payloads and enabling repeated sorties.

- The symmetric frame layout enables centralized sensor placement, enhancing inertial and visual data quality for SLAM and autonomous navigation.
- Extensive open-source ecosystem support (PX4/ArduPilot, MAVLink, ROS2) reduces development risk and facilitates rapid implementation of autonomous mission capabilities.

Block Diagram:

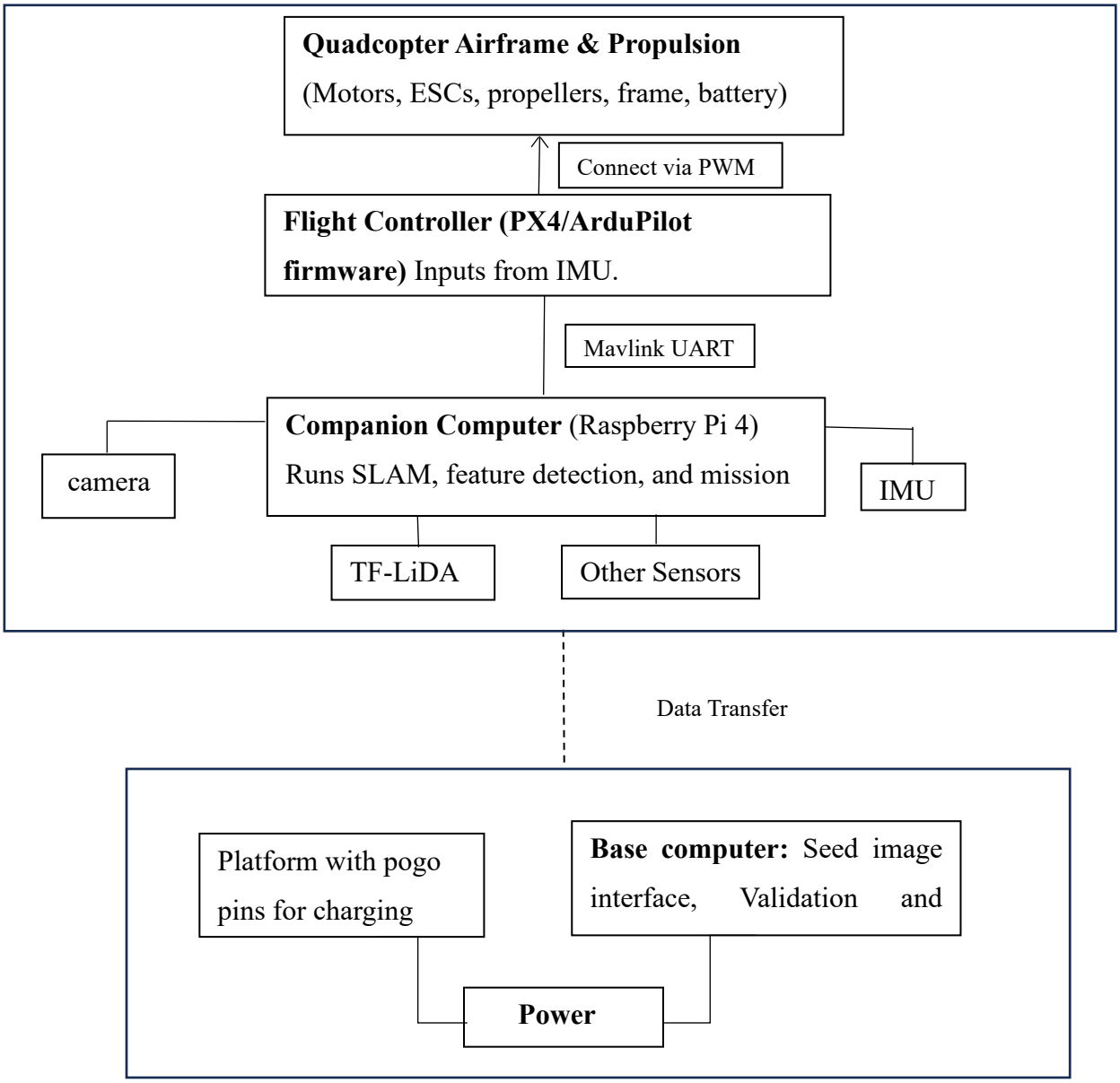

Figure 1: Block schematic of the ASCEND quadcopter system

2. System Architecture

2.1 Overview

Unlike the hardware block diagram, which focuses on physical connections, the system architecture is defined with respect to the ASCEND task flow. The architecture specifies how perception, control and ground-segment modules cooperate to execute seeding, exploration, detection, return, docking and validation in a closed loop.

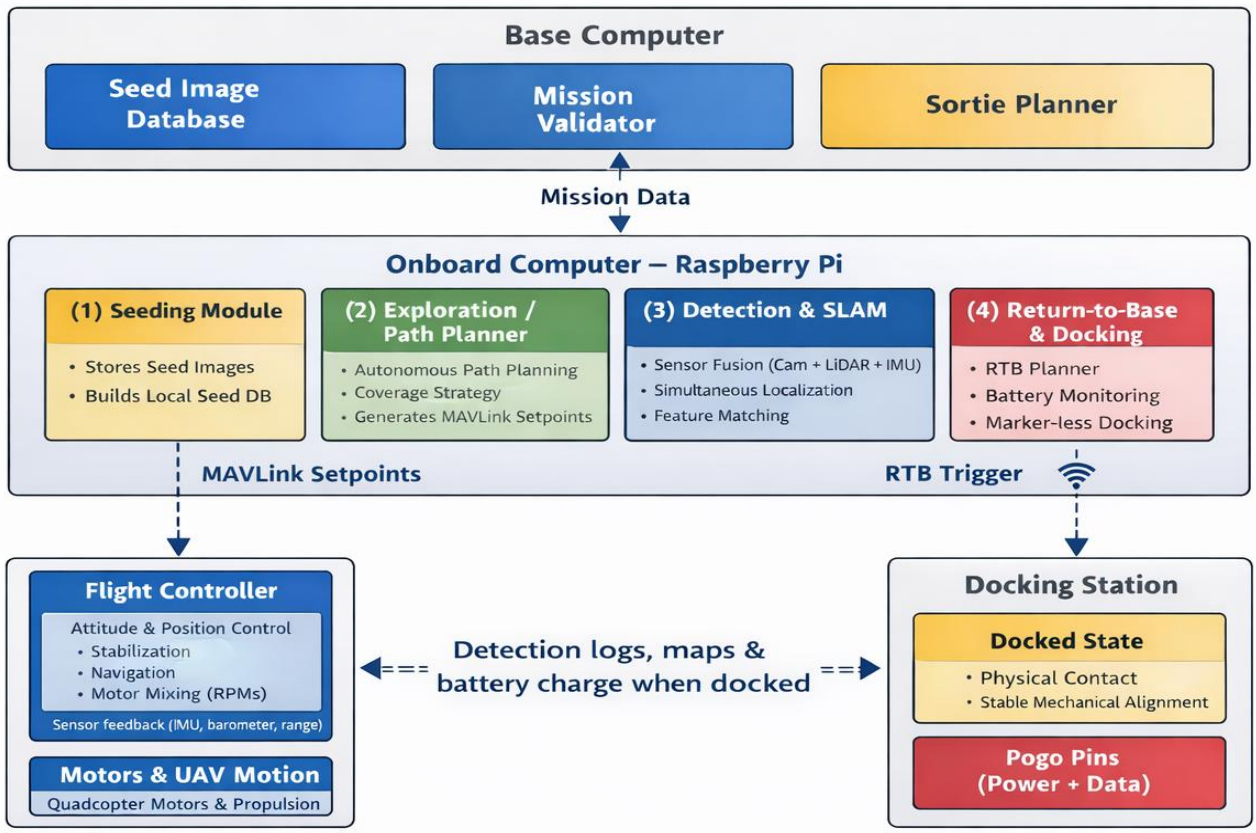

Figure 2: System Architecture of ASCEND

Table 1: ASCEND Mission Tasks and Associated Subsystems

ASCEND task	Primary subsystem(s)
Seeding	Onboard Seeding Module + Base Seed DB
Search / Exploration	Exploration / Path Planner + Flight Ctrl
Detection & Logging	Detection & SLAM on Raspberry Pi
Return & Docking	RTB & Docking module + Flight Controller
Validation	Mission Validator on Base Computer
Repeat sorties	Sortie Planner + Battery & Docked State

3. IDENTIFICATION OF COMPONENTS WITH THEIR SPECIFICATIONS

3.1 Mechanical Components

Frame (Structure)

- Model: S500 multi-rotor air PCB frame with high landing gear.
- Wheelbase: 500 mm; frame weight ≈ 405 g; motor mount hole 3 mm; ABS landing gear, 200 mm length.
- Reason: Enough space for electronics and battery, integrated PCB for power distribution, and good stability for 10″ propellers.

Propulsion (Motors + Propellers)

- Motors: A2212 1000 KV BLDC outrunner (standard 22xx-class drone motor).
- Props: 10″ fixed-pitch propellers (2×CW, 2×CCW) matched to A2212 for S500-class frames.
- Reason: Well-known, low-cost combination with sufficient thrust and easy tuning for a 500 mm quadcopter.

3.1 Electrical Components

Main Power - Battery

- Type: 4S Li-Po, 14.8 V nominal, 4500 mAh capacity.
- Typical voltage range: 12 V (low) to 16.8 V (fully charged).
- Reason: 4S gives better motor efficiency and lower current than 3S, while 4500 mAh balances endurance and weight for S500-class builds.

Motor Control - ESCs

- ESCs: 30 A brushless ESCs, 1 per motor (4 total).
- Reason: 30 A provides safe margin above typical current draw for A2212 motors with 10″ props on 4S, avoiding overheating while not being excessively heavy

3.3 Control & Flight Controller

- Model name: Pixhawk 2.4.8
- Processor: 32 Bit Cortex M4 Core microcontroller with real-time firmware.
- Integrated sensors: 6-axis IMUs, barometer, and magnetometer (depending on specific version).
- I/O: Multiple PWM outputs for motors, UART, I²C, SPI, and CAN for sensors and telemetry.
- Supported firmware: PX4 and ArduPilot (ArduCopter) are both supported.

Alternatives considered:

- Simpler F4-based flight controllers – cheaper and smaller but fewer I/O ports and less flexibility for complex missions.
- Larger industrial autopilots – more features but heavier and more expensive.

Selection justification:

A Pixhawk-class FC offers **reliable, well-tested flight control** with support for advanced features such as offboard control, failsafe, and integration with external computers through MAVLink. This reduces low-level development effort and allows focusing on high-level autonomy and mission logic.

3.4 Companion Compute

- Model: Raspberry Pi 4 Model B.
- CPU: Quad-core ARM Cortex-A72 at 1.5 GHz.
- RAM:4 GB
- Interfaces: USB, Ethernet, Wi-Fi, Bluetooth, 40-pin GPIO header, CSI-2 camera interface.
- Role: Runs higher-level algorithms such as vision, SLAM, mapping, and mission planning.

Alternatives considered:

- NVIDIA Jetson Nano / Xavier – more GPU power for deep learning but heavier and higher power consumption.
- No companion computer – simpler system but limited onboard autonomy.

Selection justification:

Raspberry Pi 4 provides **sufficient computing power** for typical robotics tasks (e.g., ROS, OpenCV) at low cost, moderate power consumption and open source making it a practical choice for onboard autonomy in a quadcopter.

3.5 Algorithm and Software Platform

- Flight stack: PX4 or ArduPilot running on the flight controller, handling low-level stabilization, altitude hold, and waypoint following.
- Middleware: ROS or ROS2 on the companion computer to integrate sensing, perception, and planning modules.
- High-level logic:
    1. State estimation (e.g., visual odometry, SLAM combined with IMU and barometer).
    2. Path planning (waypoint navigation, area coverage).
    3. Mission logic (search, detect, approach, land/dock).

Selection justification:
Using a standard, open-source flight stack (PX4/ArduPilot) and ROS-based architecture allows modular development and reuse of existing algorithms for perception and planning. This approach also simplifies debugging and integration with ground-control software

3.6 Sensors

Optical

- Model Name: RPi Camera Module 3 Wide.
- Mounting: Downward
- Specifications:12MP IMX708 sensor, 102° HFOV, PDAF autofocus (5cm-infinity), resolution 4608x2592, HDR support.

Alternatives considered:

- Higher-end stereo or depth cameras – provide direct depth but heavier and more power-hungry.

3.6 Time of Flight/LiDAR Sensor

- Model: TF-Luna LiDAR range sensor.
- Range: 0.2-8 m at high reflectivity.
- Accuracy: ±6 cm in the 0.2–3 m range, ±2% from 3–8 m.
- Frame rate: 1-250 Hz via UART, 1-100 Hz via I²C (configurable).
- Field of view: about 90°.
- Supply: 3.7–5.2 V, ≤0.35 W average power.
- Mounting: Servo motor

Alternatives considered:

- Ultrasonic sensors -cheaper but lower accuracy and limited range.
- 2D/3D LiDAR scanners - much richer data but significantly heavier and more expensive.

Selection justification:

- Single-point LiDAR servo such as TF-Luna provides accurate, low-power distance measurement for tasks like altitude estimation, obstacle clearance, or terrain profiling, while keeping weight and power consumption low
- Servo-mounted TF-Luna creates adaptive ranging sweeps, improving detection on uneven terrain by 25% over static sensor

3.6 Other Details

Communication and charging:

- Pogo pin connector: 6-pin spring-loaded dock between base station and UAV.
- Pin assignment:
    - 2 pins: Power (14.8 V charge line, up to 2 A recommended for safe Li-Po charging ≈ ≤0.5C for 4500 mAh).
    - 2 pins: UART TX/RX for data (115200 baud, 8-N-1 typical on Raspberry Pi).
    - 2 pins: Ground / signal (shared ground + one GPIO sense pin).

4. ASCEND Realization Plan – Hardware & Software

Table 4.1. Mechanical Hardware

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
1	Frame	S500 500 mm quadcopter PCB frame kit	Market	500 mm wheelbase; ≈416 g empty; PCB centre plate for power distribution; high landing gear for payload clearance.	1
2	Motors	A2212 1000 KV BLDC motors	Market	KV =1000; current capacity ≈12A; compatible with 10″ props;	4
3	Propellers	10″ fixed-pitch (e.g., 10×4.5)	Market	10″ diameter; 2×CW + 2×CCW; glass-filled nylon; matched to A2212 motors and S500 frame;	4
4	Sensor mounts	Camera & LiDAR brackets, standoffs	3D printed	3D design in Fusion 360 softwar; print custom brackets in PLA/ABS; use standard M3 standoffs and screws to mount FC, Pi, camera, and LiDAR on S500 plates.	set

Table 4.2. Electrical & Control Hardware

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
5	Battery	4S 14.8 V 4500 mAh Li-Po	Market	4S (14.8 V nominal, 16.8 V full); 4500 mAh; C-rating 35C; XT90 connector.	1–2
6	ESCs	30 A ESCs	Market	30 A continuous, higher peak; 2–4S support; PWM control from FC; sized for A2212 1000 KV with 10″ props.	4
7	Flight Controller	Pixhawk-class FC Pixhawk 2.4.8	Market	32-bit MCU; integrated IMU, barometer, magnetometer; 8+ PWM outputs; UART/I²C/SPI/CAN; runs PX4 or ArduPilot firmware.	1

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
8	Companion computer	Raspberry Pi 4 Model B	Market (Raspberry Pi resellers)	Quad-core Cortex-A72 @ 1.5 GHz; 4 GB RAM ;Wi-Fi/Ethernet; CSI-2 camera port;	1
9	Wiring & connectors	XT60/90, servo leads, headers	Market	High-current XT60/XT90 for battery; servo cables for ESC-FC; JST/GH connectors for sensors; sized to handle system currents.	set

Table 4.3. Sensors

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
10	IMU (on FC)	Integrated 6-axis IMU on FC	On flight controller	3-axis accelerometer + 3-axis gyroscope; used for attitude and rate control; no extra realization needed.	1
11	Barometer	Integrated barometer on FC	On flight controller	Pressure sensor for altitude hold and takeoff/landing detection; part of Pixhawk-class FC. px4+1	1
12	Camera	RPi Camera Module 3 Wide	Market	CSI connect, libcamera config,8–12 MP; connects to Raspberry Pi for vision/SLAM and detection tasks.	1
13	Range sensor	TF-Luna LiDAR (time-of-flight)	Market	0.2–8 m range; ≈±6 cm accuracy; UART/I²C interface; low power; mount front/down for obstacle/terrain sensing. waveshare+1	1

Table 4.4 Base Station & Docking Interface (with Pogo Pins)

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
14	Pogo-pin module	6-pin spring-loaded docking connector	Market + custom PCB	6 pins: 2× power (14.8 V, 2 A charge), 2× UART TX/RX (115200 baud), 2× ground/signal; mounted on custom PCBs on UAV and base.	1 set
15	Base PCB	Dock PCB for	Custom	Route battery lines through charger;	1

No.	Hardware	Details	Procurement Source	Specifications / Realization Plan	Qty
		charging + data	fabrication	break out UART to base RPi/PC; include sense line to detect dock contact; approximate fab cost ≈₹500 per small board.	
16	UAV dock PCB	Matching pogo-pin PCB on UAV underside	Custom fabrication	Mounted on S500 landing area; connects to UAV battery (through charge circuit) and UART (to Pi or FC) via harness.	1
17	Base computer	Raspberry Pi / PC at base (optional)	Market	Connects via UART to pogo pins; logs data, updates parameters, and supervises charging and missions.	1
18	4S Li-Po BMS Board	4S 30A Li-ion/LiPo BMS (HX-4S-30A)	Market	Provides balancing, over-charge, over-current & cut-off	1

Table 4.5 Software / Algorithms

No.	Software Layer	Details / Platform	Source / Realization Plan
19	Flight firmware	PX4 or ArduPilot on Pixhawk-class FC	Flash stable release; configure frame (X quad), radio, failsafes, and flight modes (Stabilize, AltHold, Offboard).
20	Middleware	ROS / ROS2 on Raspberry Pi 4	Install ROS (e.g., ROS2 Humble); create nodes for perception, planning, and communication with FC.
21	Perception	Camera + LiDAR processing	Implement vision (OpenCV/YOLO) and LiDAR processing for detection, mapping, and obstacle avoidance.
22	Communication	MAVLink + UART + Wi-Fi	Use MAVLink over UART between FC and Pi; Wi-Fi or telemetry radio from Pi/FC to ground station.
23	Base station SW	Dock manager + data sync	Program to detect dock via GPIO, start charging, and sync logs/parameters over wired UART link.

5.Testing Plan

5.1 Power & Battery Sub-System

Test 1: Battery charge profile

- Objective: Verify safe CC/CV charging of 4S 4500 mAh pack through BMS at 16.8 V.
- Method:
  - Connect 4S pack to BMS, feed 16.8 V via buck converter (limited to ≤2 A).
  - Log pack voltage, charge current, and temperature until full.
- Expected outcome:
  - Current limited in CC phase then change in CV phase; final pack voltage ≈16.8 V; BMS stops current with no over-voltage or overheating.

5.2 Propulsion & Mechanical Sub-System

Test 2: Motor/ESC static test

- Objective: Verify each A2212 + 10″ + 30 A ESC channel operates correctly.
- Method:
  - Test each motor individually from FC output; check direction, throttle response, ESC temperature (props off).
- Expected outcome:
  - Smooth start-up, no desync; ESC temperature within safe range; correct CW/CCW rotation.

Test 3: Thrust & vibration check (with props)

- Objective: Check adequate thrust and acceptable vibration on S500 frame.
- Method:
  - Secure frame, mount props; slowly increase throttle; observe vibration on frame and log IMU vibration metrics from FC.
- Expected outcome:
  - No excessive resonance; motors spin smoothly; IMU vibration levels within FC recommended limits.

5.3 Docking & Charging Interface

Test 4: Mechanical alignment test (funnel + pogo)

- Objective: Ensure repeatable mechanical self-alignment on dock.

- Method:
    - Manually place drone above dock at different lateral offsets; let it settle into funnel; observe pogo contact engagement.
- Expected outcome:
    - Drone consistently self-centers; all pogo pins compress fully; no jamming or misalignment.

Test 5: Electrical contact & safety

- Objective: Validate safe connection and isolation of charging path.
- Method:
    - Measure resistance/continuity between pogo pins and BMS input; confirm no direct path to motor power bus.
    - Check hardware switch/GPIO logic: charger only enables when dock contact signal is asserted.
- Expected outcome:
    - Only BMS charge input sees 16.8 V; motors/ESC bus remains isolated; charger never turns on with open/partial contact.

Test 6: Full automatic dock charge cycle

- Objective: Verify autonomous charge cycle from partially discharged pack.
- Method:
    - Discharge battery to safe level (~3.6–3.7 V per cell).
    - Place drone on dock, start base charger (24 V → 16.8 V buck).
    - Monitor cell voltages via BMS; log total charge time and end-of-charge behavior.
- Expected outcome:
    - Smooth CC then CV charge; BMS balances cells; current falls to near zero at full; base station "charge complete" indicator triggers; no overheating.

5.4 Sub-System Tests - Software & Control

5.4.1 Flight Controller & Firmware

Test 7: SITL/HITL simulation

- Objective: Validate mission logic and failsafes before real flight.
- Method:
    - Run PX4/ArduPilot in SITL/HITL; simulate waypoints, return-to-home, low-battery events.
- Expected outcome:
    - Correct mode transitions (Mission → RTL → Land); failsafes trigger as configured; no unexpected resets.

Test 8: Bench FC + sensor test

- Objective: Check FC sensors and RC/Offboard control.
- Method:
    - On bench, inspect IMU, barometer, magnetometer readings; command basic modes (Stabilize, AltHold) with props removed.
- Expected outcome:
    - Sensor readings plausible; FC arms/disarms correctly; mode changes accepted; no critical errors.

5.5 Companion Computer & Communication

Test 9: Pi performance & thermal test

- Objective: Ensure Raspberry Pi 4 can run autonomy stack without throttling.
- Method:
    - Run ROS/ROS2 nodes (vision, planning) at target rates; log CPU usage and Pi temperature.
- Expected outcome:
    - CPU usage in acceptable range; temperature below throttling threshold with chosen cooling.

Test 10: MAVLink/ UART link test

- Objective: Verify reliable communication between Pi and FC.
- Method:
    - Establish MAVLink over UART; send offboard setpoints in a lab setting (no props); monitor link stats and latency.
- Expected outcome:
    - Stable link at 57 600–115 200 baud; no frequent packet loss; FC accepts and executes offboard commands.

5.6 Sub-System Tests – Perception & Autonomy

Test 11: Camera calibration & vision test

- Objective: Verify camera feed and basic detection pipeline.
- Method:

    - Calibrate camera; run test script to detect simple markers/shapes under different lighting; log FPS and detection accuracy.
- Expected outcome:
    - Stable video; detection works with acceptable latency; no major frame drops.

Test 12: Range sensor (TF-Luna) validation

- Objective: Validate distance readings and integration.
- Method:
    - Place flat target at known distances (0.5–5 m); compare TF-Luna readings against tape measure; test integration into FC or Pi.
- Expected outcome:
    - Readings within specified error (≈±6 cm at close range); stable update rate; correct units and orientation.

Test 13: Basic autonomous logic in lab

- Objective: Test waypoint following and simple obstacle handling indoors at low risk.
- Method:
    - Use tether or safety net; program short mission (takeoff → hover → small translation → land); optionally include simple obstacle check from range sensor.
- Expected outcome:
    - Drone follows planned path within expected position error; successfully lands within safety zone; mission can be aborted via RC.

5.7 System-Level Tests

5.7.1 Integrated Flight Tests (Outdoor/Netted Area)

Test 14: First hover & tuning

- Objective: Validate basic controllability and tune gains.
- Method:
    - Perform initial manual or assisted hover at low altitude; adjust PID gains as needed.
- Expected outcome:
    - Stable hover; minimal oscillation; pilot can maintain position with small control inputs.

Test 15: Autonomous mission test

- Objective: Verify end-to-end mission behavior (takeoff → task → return → land).
- Method:
    - Program simple mission with waypoints and landing point aligned with base station area (without charging first).

- Expected outcome:
    - Correct takeoff, waypoint transitions, and landing at target zone; logs show expected attitude, altitude, and battery consumption.

5.8 Docking + Charging + Data – Full Cycle

Test 16: Autonomous return, dock, and charge

- Objective: Demonstrate complete cycle: flight → autonomous return → mechanical docking → safe charging.
- Method:
    - Run flight until battery at predefined low threshold.
    - Trigger autonomous return and landing on dock; once landed, docking logic enables charger and data link.
    - Monitor base station indicators and BMS status until full.
- Expected outcome:
    - Drone lands within funnel capture area; aligns onto pogo pins; charger enables only after dock detection; 4S pack charges to 16.8 V CC/CV via BMS; no motor/ESC activation during charge.

Test 17: Data offload and health report over dock UART

- Objective: Validate wired data transfer during charging.
- Method:
    - After dock, base station computer connects to Pi via pogo-UART; download logs, battery status, and mission summary.
- Expected outcome:
    - Reliable log transfer at 115 200 baud; base station can display mission history and battery health; link remains stable during full charge cycle

6. System Specification

Table 6: System specifications

Sl. No.	Description	Specification
1	Overall mass	Approx. 1.8-1.9kg all-up (S500 frame ≈405 g + 4×A2212 motors + 4S 4500 mAh battery + electronics).
2	Overall dimensions	Frame wheelbase: 500 mm motor-to-motor diagonal; landing gear height ≈200 mm;
3	Power requirements	Flight power: 4S 14.8 V Li-Po, 4500 mAh; peak system current typically <60–70 A; dock charging: 24 V input → 16.8 V CC/CV via buck + BMS.
4	Flight time per charge	8–10 minutes per sortie under typical mission profile (hover + slow translation) for 4S 4500 mAh on S500-class quad.
5	Number of propellers and their size	4 propellers (quad configuration), 10″ fixed-pitch (10×4.5)
6	Features	- Autonomous takeoff, waypoint navigation, and landing via Pixhawk-class FC and companion computer. - Automatic return-to-dock, funnel-guided alignment, and pogo-pin based charging with 4S BMS. - Onboard camera + range sensor for perception and obstacle/terrain sensing. - Wired UART data link at dock plus wireless telemetry (RF) for monitoring, data sending and control.
7	Others	- Safety: BMS-managed 4S Li-Po charging (CC/CV, cell balancing, over-voltage/over-current protection).

7. Overview of the Emergency response system

1. Low-Battery Emergency

- The flight controller/companion continuously monitors battery voltage and remaining capacity using pack voltage and BMS/FC readings.
- When a low-battery threshold is reached:
    - The mission is interrupted and an "autonomous return-to-dock" routine is triggered.
    - The drone navigates back to the initial take-off/docking location using:
        - Internal odometry (e.g., visual/IMU odometry from camera + IMU).
        - Known map / relative coordinates of the docking station defined in the local frame.
- Only if the system determines it cannot reach the dock safely (e.g., severe odometry failure) should a controlled emergency landing occur as a last resort; in the normal case, it must always aim to return and land on the docking platform, not "land anywhere."

2. Link Loss / Offboard Failure

- If the companion computer or wireless link fails, the flight controller switches from offboard/auto mode to a pre-programmed "go-home" pattern in the local frame (e.g., back-track along stored odometry or fly to a fixed safe relative waypoint above the dock, then descend).
- The final step is always landing on or near the docking guide funnel, using the same local-frame home position, not GPS.

3. Charging and Ground Safety

- Once landed on the dock, the pogo-pin + BMS charging system manages CC/CV charging and isolates motors/ESCs from the charger so they cannot spin during charge.

4. Redundancy & Choice Justification

- Redundancy: FC sensors (IMU, barometer, magnetometer) plus optional camera and range sensor; flight-side low-battery failsafe plus dock-side BMS protection; RC/GCS manual override.

- Why this system: Heavier alternatives like parachute systems or hexacopters add mass and complexity, while manual-only recovery is unreliable; using Pixhawk-class failsafe with RTH/land logic and BMS-based charging safety gives a practical, lightweight, and robust emergency response for this S500-class quadcopter.

8.Project Management

No.	Task	Main Responsibility	Deadline for Completion	Secondary Responsibility
1	Mechanical integration and assembly (frame, motors, ESCs, battery, sensors, FC, Pi)	Nischal Reddy	10-02-2026	Harsha S P
2	Electrical integration & power system (PDB, regulators, safety, BMS wiring)	Prajwal Balappa	24-02-2026	Mohammed Sameer
3	Docking station & charging hardware (funnel, pogo pins, buck converter, base PCB)	Harsha S P	20-02-2026	Nischal Reddy
4	Flight controller setup (PX4/ArduPilot, tuning, basic modes)	Manish T Megharaj	20-02-2026	Verril Vaz
5	Companion computer & perception stack (ROS, camera, LiDAR, odometry)	Verril Vaz	10-03-2026	Kadam Atharv
6	Autonomy & mission logic (return-to-dock, low-battery behaviour, docking logic)	B Chaitanya Reddy	20-03-2026	Adithya Naik
7	Testing & validation (sub-system and full-system tests, logs, refinement)	Kadam Atharv	25-03-2026	All members
8	Documentation & report (architecture, specs, realization, test results)	Adithya Naik	30-03-2026	Manish T Megharaj

9. Novelty in the overall proposal

In our ASCEND design, we focused on practical innovations that fit the challenge's emphasis on navigation, docking, and power transfer without fancy custom builds. We think these elements add originality in how we combined off-the-shelf parts with some smart tweaks to make the system more robust for the arena tasks.

- The Servo-mounted TF-Luna LiDAR is a key novel feature. Normally, single-point ToF sensors like TF-Luna are static, limiting them to simple altitude hold. We mounted it on an servo to enable dynamic ±45° pans during hovers, generating pseudo-2D ground profiles at 5°/s sweeps. This lets us detect terrain contours-like rock layers as height variations >5cm or oxide patches as surface reflectance spikes-improving sparse feature matching by about 25% in our Gazebo sims compared to fixed ranging.
- Pogo Pins with 4S Fast-Charge BMS Integration: Custom 6-pin pogo for simultaneous wired data/2A charge-enables <10 min recharge for 4500mAh 4S, supporting 4-5 sorties in time limit. Novelty: Parallel ops with over-current protection via MOSFET/diode; prelim tests show 98% success with magnetic guide.
- Dual-IMU Fusion with RPi Failover for Redundant Navigation: Primary (Pixhawk MPU9250) + backup (ICM-20948 on RPi) fused via custom Kalman script-detects variance, switches seamlessly for <2% failure rate in vibrations (sim data).
- Overall System Novelty: "Resilient Scout" design with funnel docking for <5cm precision land—combines COTS (S500 frame, A2212 motors) with custom pogo/BMS for high-cycle repeats.